

CATION DISTRIBUTION IN Eu-SC-Fe GARNETS

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The optimal formula to calculate the lattice constant a of garnet compounds based on their cation distribution is derived from an equation that relates the cubic lattice constant with the average ionic radii at different coordination numbers, as described in the paper.

The equation used by Stročka et al. [27] for this purpose is:

$$a = b_1 + b_2 r_{V1}^{6/3} + b_3 r_{V1}^4 + b_5 r_{V1}^2 + b_6 r_{V1} r_{V2}$$

where b_i ($i = 1, 2, \dots, 6$) are constants, and r_{V1} , r_{V2} are the average radii for octahedral and tetrahedral sites, respectively. The equation is:

$$a = b_1 + b_2 r_{V1}^{6/3} + b_3 r_{V1}^4 + b_5 r_{V1}^2 + b_6 r_{V1} r_{V2}$$

The values of b_i are determined by fitting the equation to experimental lattice constants for a set of garnet compounds with known cation distributions. The ionic radii used in this equation, such as r_{V1} and r_{V2} , refer to the average radii at octahedral (V1) and tetrahedral (V2) coordination numbers.

The paper also notes that for the specific case of Eu-Sc-Fe garnets, assuming the change in ionic radius with coordination number is similar to that of Eu^{3+} and Sc^{3+} , one can estimate r_{V2} :

$$r_{V2} = r_{\text{Eu}^{3+}} - (r_{\text{Eu}^{3+}} - r_{\text{Sc}^{3+}})$$

where $r_{\text{Eu}^{3+}}$ is the ionic radius for Eu^{3+} in eight-fold coordination, and $r_{\text{Sc}^{3+}}$ is the ionic radius for Sc^{3+} in eight-fold coordination. The value of r_{V2} thus obtained is used to calculate the lattice constant.

The equation (6) from Stročka et al. [27] is:

$$a = b_1 + b_2 r_{V1}^{6/3} + b_3 r_{V1}^4 + b_5 r_{V1}^2 + b_6 r_{V1} r_{V2}$$

This equation, when applied with the appropriate values of b_i and ionic radii for Eu3+ and Sc3+, provides a good fit to experimental lattice constants for garnet compounds. The paper uses this equation to determine the cation distribution in Eu-Sc-Fe garnets by comparing calculated lattice constants (a_{th}) with experimental values (a_{exp}), as shown in Table 1.

The best agreement between a_{exp} and a_{th} is obtained for a particular cation distribution, which is then used to infer the actual composition of the samples.